

ICP Final Project Report

```
#include<stdio.h>
int main(){

    printf("My C Programming Project: ");

    return 0;
}
```

Project Title:

Student Scholarship Eligibility Checker

Submitted By:

Name:	Faiza Batool.
ID:	242357.
Section:	8.

Submitted To:

Instructor's Name:	Bibi Sara.
Course:	ICP
Semester:	Fall 2026
Academic Year:	2026
Submission Date:	August 9, 2026

\$ █

1. Abstract:

Choosing the right scholarship can be difficult for students because every country and scholarship program has different requirements. This project was created to make this process easier by checking a student's information against the scholarship requirements.

This c program checks a student's eligibility for scholarships in 10 different countries. The program asks for information such as SSC/O-Level and HSSC/A-Level, marks, age, IELTS score, and family income. It then compares this information with the requirements and shows whether the student qualifies for a fully funded scholarship, partially funded scholarship, or no scholarship.

The project also shows the reasons when a student does not meet the requirements. The reason for the rejection may help the students to work on their weakness. Through this project, I practiced important C programming concepts such as variables, input and output, conditional statements, logical operators, functions, strings, and basic program control.

2. Introduction & Background:

Scholarships are a good way for students to continue their education, especially when the cost of studying is high. However, different scholarships and countries have different requirements. Students may need to meet certain conditions related to their academic marks, age, IELTS score, and family income.

To make this process easier, I developed a Scholarship Eligibility System using the C programming language. The system allows students to select their favourite place to study, firstly it allows them to select the continent and after that a country and asks them to enter their personal and academic information. It then checks the information according to the requirements set for that country. The system covers 10 countries from four different regions: Asia, Europe, Oceania, and North America. Each country has requirements for both fully funded and partially funded scholarships.

The program uses different requirements for each country and scholarship type. These requirements include SSC marks, HSSC marks, maximum age, maximum family income, and IELTS score.

The following table shows all the scholarship requirements used in the program.

Continent	Country	Scholarship	SSC	HSSC	Max Age	Max income	IELT S
Asia	China	Fully Funded	85%	85%	25	\$2,000	6.0

Asia	China	Partially Funded	75%	75%	25	\$3,000	6.0
Asia	Japan	Fully Funded	90%	90%	22	\$1,500	6.5
Asia	Japan	Partially Funded	85%	85%	22	\$2,000	6.0
Asia	Turkey	Fully Funded	80%	80%	25	\$2,500	6.0
Asia	Turkey	Partially Funded	70%	70%	25	\$3,500	5.5
Oceania	Australia	Fully Funded	90%	90%	21	\$1,300	6.5
Oceania	Australia	Partially Funded	85%	85%	21	\$1,800	6.0
Oceania	New Zealand	Fully Funded	85%	85%	21	\$1,500	6.5
Oceania	New Zealand	Partially Funded	80%	80%	21	\$1,500	6.0
Europe	Germany	Fully Funded	90%	90%	23	\$2,000	6.5
Europe	Germany	Partially Funded	80%	80%	23	\$3,000	6.0
Europe	France	Fully Funded	90%	90%	21	\$2,200	6.5
Europe	France	Partially Funded	85%	85%	22	\$2,000	6.0
Europe	Italy	Fully Funded	80%	80%	25	\$1,500	6.0
Europe	Italy	Partially Funded	70%	70%	25	\$1,800	5.5
North America	Canada	Fully Funded	90%	90%	21	\$3,000	6.5
North America	Canada	Partially Funded	85%	85%	21	\$3,500	6.0

North America	USA	Fully Funded	90%	90%	22	\$2,500	6.5
North America	USA	Partially Funded	85%	85%	22	\$2,000	6.0

The program compares the student's information with these requirements. Based on the result, the student can qualify for a fully funded scholarship, a partially funded scholarship, or be rejected.

3. Objectives:

The main objectives of this project are:

- To create a simple program for checking scholarship eligibility.
- To provide fully funded and partially funded scholarship results.
- To show the reason when a student is not eligible.
- To practice C programming concepts.
- To improve problem-solving and logical thinking skills.
- To apply C programming to a real-world problem.

4. Implementation:

This program follows a simple process. First, the user selects a continent and a country. The program then asks the user to enter their personal and academic information, including their name, SSC marks, HSSC marks, age, IELTS score, and family income.

After receiving the information, the program compares it with the requirements for the selected country. If the student meets the requirements for a fully funded scholarship, the program displays that result. If the student does not qualify for fully funded but meets the requirements for partially funded, the program displays the partially funded result. If the student does not meet the minimum requirements, the program displays a rejection message along with the reasons for rejection.

5. Result/Output:

The program was tested using different student information and scholarship requirements. The system successfully takes the student's information, checks it against the selected country's requirements, and displays the appropriate scholarship result. The program can produce three main results,

1. If the student meets all the requirements for a fully funded scholarship, the program displays a fully funded scholarship result along with the student's information.

```

=====
Your Information :
=====
Name:                Faiza
Age:                 20
SSC result:          90
HSSC result:         89
IELTS score:         6.5
Family income:       1200
=====
Scholarship Eligibility Result :
=====

Congratulations!
You are eligible for a Fully Funded Scholarship.
Your excellence earned you the full opportunity.

```

2. If the student does not meet the requirements for a fully funded scholarship but meets the requirements for a partially funded scholarship, the program displays a partially funded scholarship result.

```

=====
Your Information :
=====
Name:                Faiza
Age:                 21
SSC result:          85
HSSC result:         86
IELTS score:         6.0
Family income:       1500
=====
Scholarship Eligibility Result :
=====

Congratulations!
You are eligible for a Partially Funded Scholarship.
A strong start to a bright future.

```

3. If the student's information does not meet the minimum requirements, the program displays a rejection result. It also shows the reasons why the student was not eligible.

=====

Your Information :

=====

Name: Faiza Batool
Age: 21
SSC result: 86
HSSC result: 80
IELTS score: 6.0
Family income: 2000

=====

Scholarship Eligiblity Result :

=====

Unfortunately you are not eligible for the scholarship:

=====

Reason(s) of Rejection:

=====

Reason: Your SSC/HSSC results do not meet the required percentage.

Activate Windows

Go to Settings to activate Windows.

=====

Your Information :

=====

Name: Faiza
Age: 26
SSC result: 67
HSSC result: 78
IELTS: No
Family income: 3500

=====

Scholarship Eligiblity Result :

=====

Unfortunately you are not eligible for the scholarship:

=====

Reason(s) of Rejection:

=====

Reason: Your SSC/HSSC results do not meet the required percentage.

Reason: Your age does not meet the required criteria.

Reason: Your family income does not meet the required criteria.

Reason: Your IELTS score does not meet the required criteria.

5. Challenges & Limitations:

1. Managing Different Countries:

One of the main challenges was managing the requirements for 10 different countries. Each country has different requirements for fully funded and partially funded scholarships. This made the code longer and required many if else conditions.

2. getchar():

Another challenge was taking the student's name after using scanf(). When I used fgets() to take the name, the program sometimes skipped the input because of the leftover newline character from scanf(). I used getchar() to remove the leftover newline before using fgets(). Understanding why this happened and how to fix it was one of the challenges I faced while developing the program.

```
char name[30];

getchar();

printf("Enter your name:\n");
fgets(name, sizeof(name), stdin);

printf("Enter your SSC/O-level marks in percentage
:(like 90)\n");
scanf("%d", &ssc);
```

3. Percentage % Input Error:

One challenge encountered was that the program only accepted marks in integer form, such as 90, but did not recognize inputs written with the percentage symbol, such as 90% ,although it did not stop the program but caused an undefined behaviour and invalid output.

4. Only for Bachelors Students:

The project is designed only for the Bs students, it does not check the eligibility for the masters or Phd.

5. Limited Numbers of Country:

The program currently includes only 10 countries. There are many other countries that offer scholarships, so the available options may not be suitable for every student.

6. Conclusion & Discussion:

This project was a useful experience for learning C programming through a real-world idea. This project can check a student's information and give a scholarship result based on the selected country's requirements. While working on the project, I learned how to connect different parts of

a program, especially conditions, and user input. I also learned that a program can become difficult to manage when many rules are added. Although the current system has some limitations, it provides a good starting point. In the future, it can be improved by adding more countries, more scholarship levels, and different English proficiency tests. Overall, this project improved my programming skills and gave me a better understanding of how to build a complete C program from the beginning.

7. Future Work

There are several ways this project can be improved in the future:

- Use arrays and pointers to store scholarship information and reduce repeated code.
- Create more functions to make the program shorter, cleaner, and easier to manage.
- Add better input validation to prevent incorrect information.
- Add scholarship options for masters and phd.
- Add more countries and scholarship programs.
- Support other English proficiency tests such as TOEFL PET and others , not only IELTS.

These improvements would make the system more organized, flexible, and useful for more students.